

Heap

Md. KINEMASTER

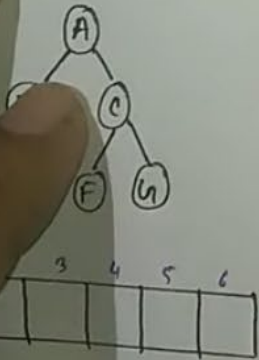
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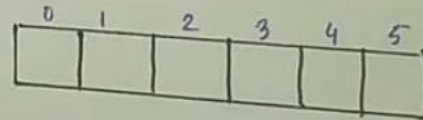
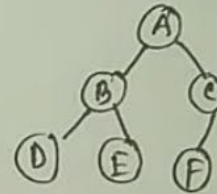
farhankhan246@gmail.com

⇒ Heap is a specialized tree-based data structure which is essentially an almost complete tree that satisfies the heap property.

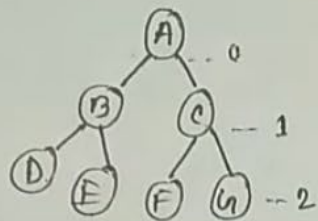
Array representation of Binary tree:



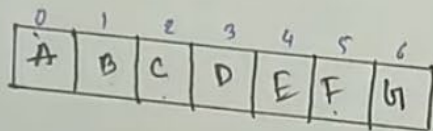
index = i



(A)



$$\begin{aligned}
 &2^{h+1} - 1 \\
 &= 2^3 - 1 \\
 &= 7
 \end{aligned}$$



$$\begin{aligned}
 &i = 1 \\
 &\text{Left } 2 \times i + 1 = 3 \\
 &\text{Right } 2 \times i + 2 = 4
 \end{aligned}$$

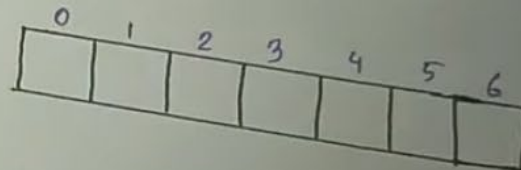
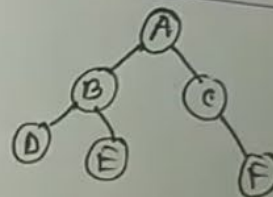
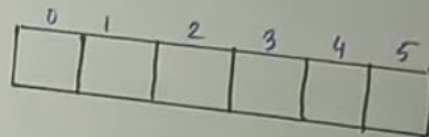
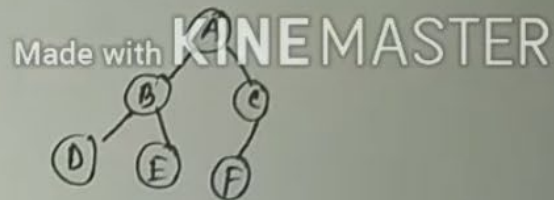
$$\begin{aligned}
 &i = 5 \\
 &5/2 = 2.5 \\
 &= 2
 \end{aligned}$$

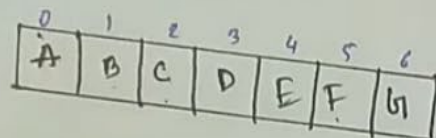
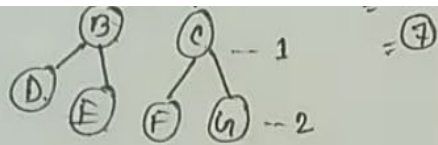
index $\Rightarrow i$

Left child $\rightarrow (2 \times i) + 1$

Right child $\rightarrow (2 \times i) + 2$

Parent $\rightarrow \lfloor (i-1)/2 \rfloor$



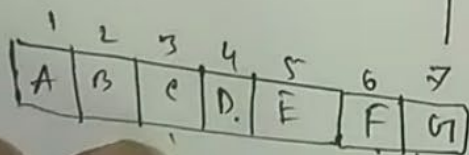


index $\Rightarrow i$

Left child $\rightarrow (2 \times i) + 1$

Right child $\rightarrow (2 \times i) + 2$

Parent $\rightarrow \lfloor (i-1)/2 \rfloor$



$$i=1$$

$$\text{Left } 2 \times 1 + 1 = \underline{\underline{3}}$$

$$\text{Right } 2 \times 1 + 2 = \underline{\underline{4}}$$

$$i=5$$

$$5/2 = 2.5 = \underline{\underline{2}}$$

Left child $\rightarrow 2 \times i$

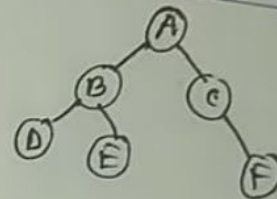
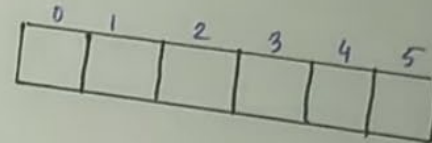
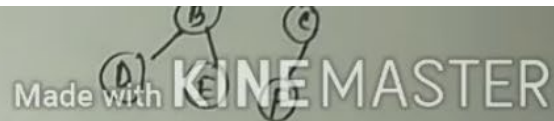
Right $\rightarrow (2 \times i) + 1$

Parent $\rightarrow \lfloor i/2 \rfloor$

$$i=2$$

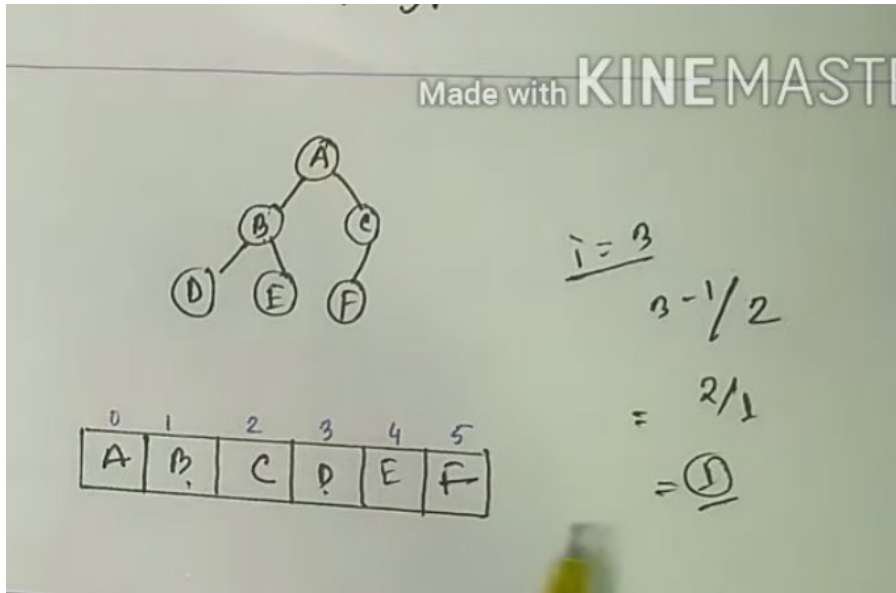
$$2 \times 2 = \underline{\underline{4}}$$

$$2 \times 2 + 1 = \underline{\underline{5}}$$

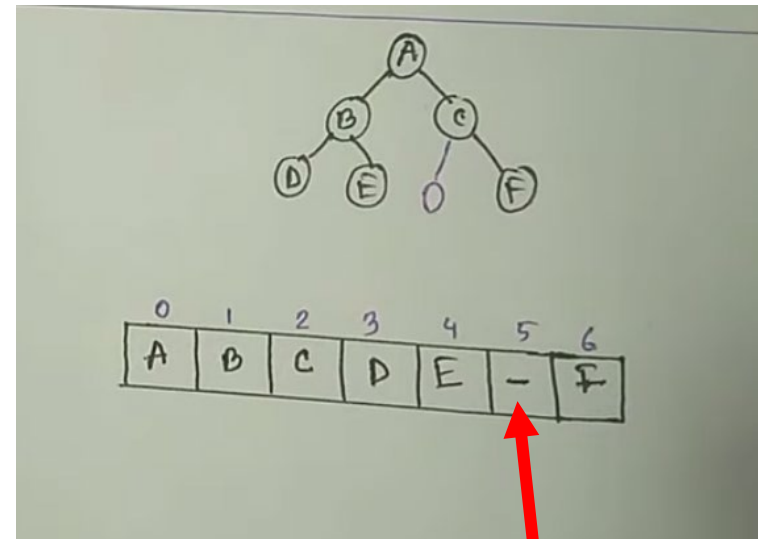


If index starts from 1
instead of 0

Complete binary tree

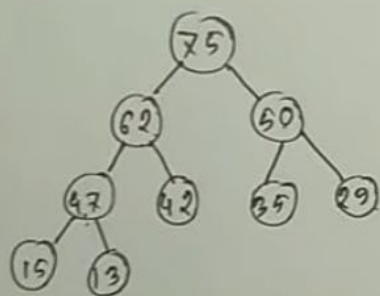


Not Complete binary tree



Because there is an empty index??

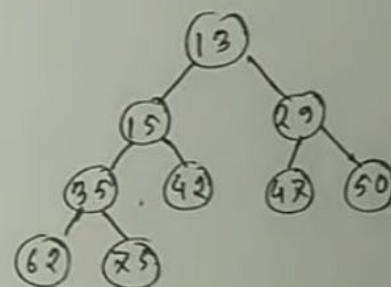
Max Heap



parent $>$ child
except root node

0	1	2	3	4	5	6	7	8
75	62	50	47	42	35	29	15	13

Min Heap



0	1	2	3	4	5	6	7	8

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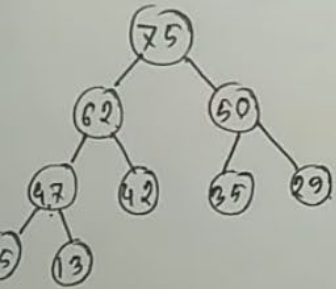
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Max Heap

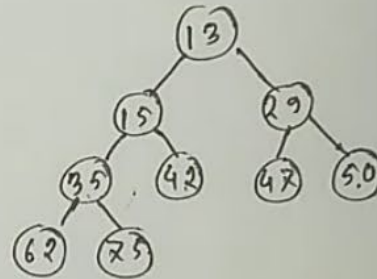
parent $\geq$ child
except root node



2	3	4	5	6	7	8
50	47	42	35	29	15	13

Min Heap

parent $\leq$ child
except root node



0	1	2	3	4	5	6	7	8
13	15	29	25	42	47	50	62	23

